

Sujith S

Software Engineer | Distributed Systems • Machine Learning • AI Systems • Full-Stack Development
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Summary

Penultimate-year Computational Statistics and Data Analytics student at VIT University (2022-2027) with a peer-reviewed publication on ML applied to financial time-series forecasting. Experienced architecting large-scale distributed systems, concurrent multi-agent pipelines, and scalable AI infrastructure across NLP, deep learning, RAG, and real-time data engineering; proficient in data structures, algorithms, concurrency, multi-threading, synchronization, and software design. Passionate about applying CS fundamentals to large-scale engineering problems in collaboration with cross-functional teams.

Education

Int.Msc Computational Statistics and Data Analytics

VIT University | 2022-2027

Relevant Coursework: Data Structures & Algorithms, Distributed Systems, Concurrency & Synchronization, Software Engineering, Computer Networks, Machine Learning, Deep Learning, NLP, Algorithmic Optimization, Statistical Modeling

CS50 Programs

Harvard University | CS50

Completed Harvard University CS50 programs including [CS50x](#), [CS50P](#), [CS50AI](#), [CS50SQL](#), and [CS50R](#) with hands-on experience in Python, SQL, backend development, algorithms, and scalable problem-solving.

Research & Publications

- **Stock Market Analysis and Prediction using Machine Learning** – [El Ruha International Conference](#), 14th Proceedings, pp. 730-740, 2025. Engineered end-to-end ML pipelines for time-series forecasting on real-world equity data; designed feature engineering stages, trained and benchmarked multiple models across market regimes. [\[Code\]](#) [\[Proceedings\]](#)

Skills

Languages: Python, Java, JavaScript, TypeScript, SQL, R

CS Foundations: Data Structures & Algorithms, Concurrency & Multi-threading, Synchronization, Distributed Systems, Software Design, Algorithmic Optimization, Performance Profiling, Debugging

AI & ML: Deep Learning, NLP, LLMs, RAG, Multi-Agent Systems, Reinforcement Learning, Statistical Inference, Time-Series Forecasting

Frameworks: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, LangChain, LangGraph, CrewAI, AutoGen, FAISS, ChromaDB, Apache Kafka, Apache Spark, MLflow, Pandas, NumPy

Tools & Infra: FastAPI, PostgreSQL, Redis, Docker, Kubernetes, Git, Linux, CI/CD, REST APIs, WebSockets, AWS, GCP

Projects

AI Workflow Automation Platform

[GitHub](#)

Python, FastAPI, AsyncIO, LangChain, LangGraph, Next.js, React, TypeScript, PostgreSQL, Redis, Celery, Docker, Prometheus, Grafana

- Architected a production-grade AI workflow platform with a DAG-based distributed execution engine supporting conditional branching, parallel execution across 10+ concurrent workflow types, fault recovery, and scheduled retries; integrated LangChain and LangGraph for multi-step AI reasoning, with real-time WebSocket monitoring, Redis caching, and Prometheus/Grafana observability across full-stack layers.

Distributed Chat System & Collaboration Platform

[GitHub](#)

FastAPI, Node.js, WebSockets, Socket.IO, Redis Pub/Sub, PostgreSQL, Next.js, TypeScript, Docker, Prometheus

- Engineered a scalable distributed real-time messaging system with sub-100ms delivery across concurrent servers; designed a Redis Pub/Sub event pipeline with cross-server broadcasting, deduplication, retry handling, and synchronization recovery, applying multi-threading and synchronization primitives to eliminate race conditions in shared-state updates across distributed nodes.

Autonomous Research Agent

[GitHub](#)

Python, LangChain, LangGraph, RAG, FAISS, ChromaDB, Hugging Face Transformers, Sentence Transformers, FastAPI, PostgreSQL, Redis

- Designed a fully autonomous research pipeline using RAG, LangGraph multi-step reasoning, and semantic vector search (FAISS, ChromaDB) to decompose complex queries, retrieve and rank sources across large unstructured corpora, and synthesize structured reports; built concurrent async execution pipelines with modular NLP preprocessing using Hugging Face Transformers for dense embedding generation and scalable knowledge extraction.

Social Media Sentiment Lakehouse

[GitHub](#)

Python, Apache Kafka, Apache Spark Streaming, Delta Lake, Hugging Face Transformers, NLTK, MLflow, PostgreSQL, Redis

- Built a large-scale real-time NLP analytics platform using concurrent Kafka producers and distributed Spark Streaming consumers for high-throughput sentiment classification and topic modeling; engineered a Bronze-Silver-Gold Delta Lake architecture with ACID compliance, synchronization guarantees for concurrent writes, and MLflow-driven automated model retraining over evolving data distributions.

Work Experience

Founder

[Shoot the Breeze](#) | 2023 - Present

- Architected and deployed AI-powered distributed systems integrating LLMs, multi-agent pipelines, and NLP infrastructure using Python and FastAPI; owned the full SDLC from system design and algorithm selection through implementation, performance profiling, concurrent execution path optimization, and production reliability evaluation.

App/Web Developer

[CropSky](#) | VIT-TBI Research & Innovation Hub | 2024 - Present

- Developed scalable backend systems and REST APIs at a university-incubated research startup; collaborated cross-functionally with research scientists and engineers to translate requirements into production-ready architecture, supporting peers in debugging and fostering a culture of shared technical ownership.

Certifications & Achievements

- [Google Generative AI Leader](#)
- [Google Cloud Computing Foundations](#)
- [FreeCodeCamp Legacy Full Stack](#)
- [OCI 2025 Certified Developer Professional](#)
- AWS [Solutions Architect](#) and [Developer](#)
- [Software Engineer Certification - HackerRank](#)